

SMART INDIA HACKATHON 2026

Hybrid QML Platform for Early Oral Cancer Detection and Extensible Medical Imaging Applications

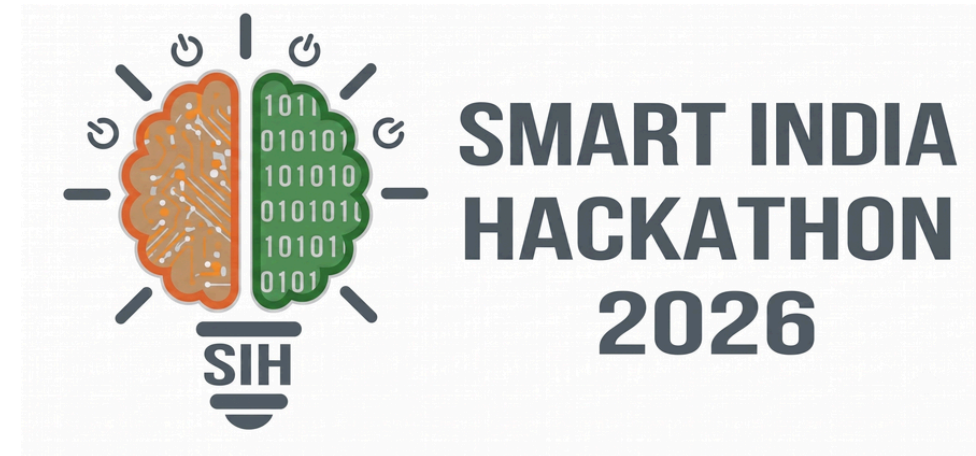
Problem Statement ID – SIH26139

Problem Statement Title - Hybrid Quantum Machine Learning Platform for Early Disease Detection

Theme - HealthTech

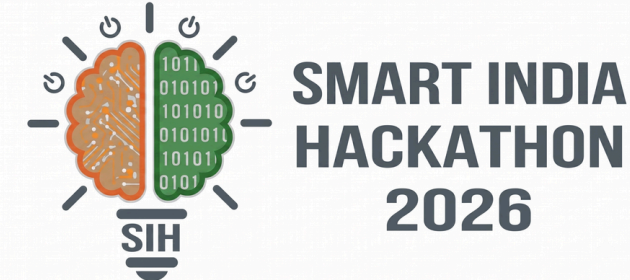
PS Category - SOFTWARE

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TEAM
BRAKET 3.1.0

Hybrid QML Platform for Early Oral Cancer Detection and Extensible Medical Imaging Applications



THE PROBLEM

Late Detection is Costly. Screening remains **difficult to access**.



~1.44 Lakh
New cases in
India (2022)



~80 thousands
Deaths in India
(2022)



Often detected
late

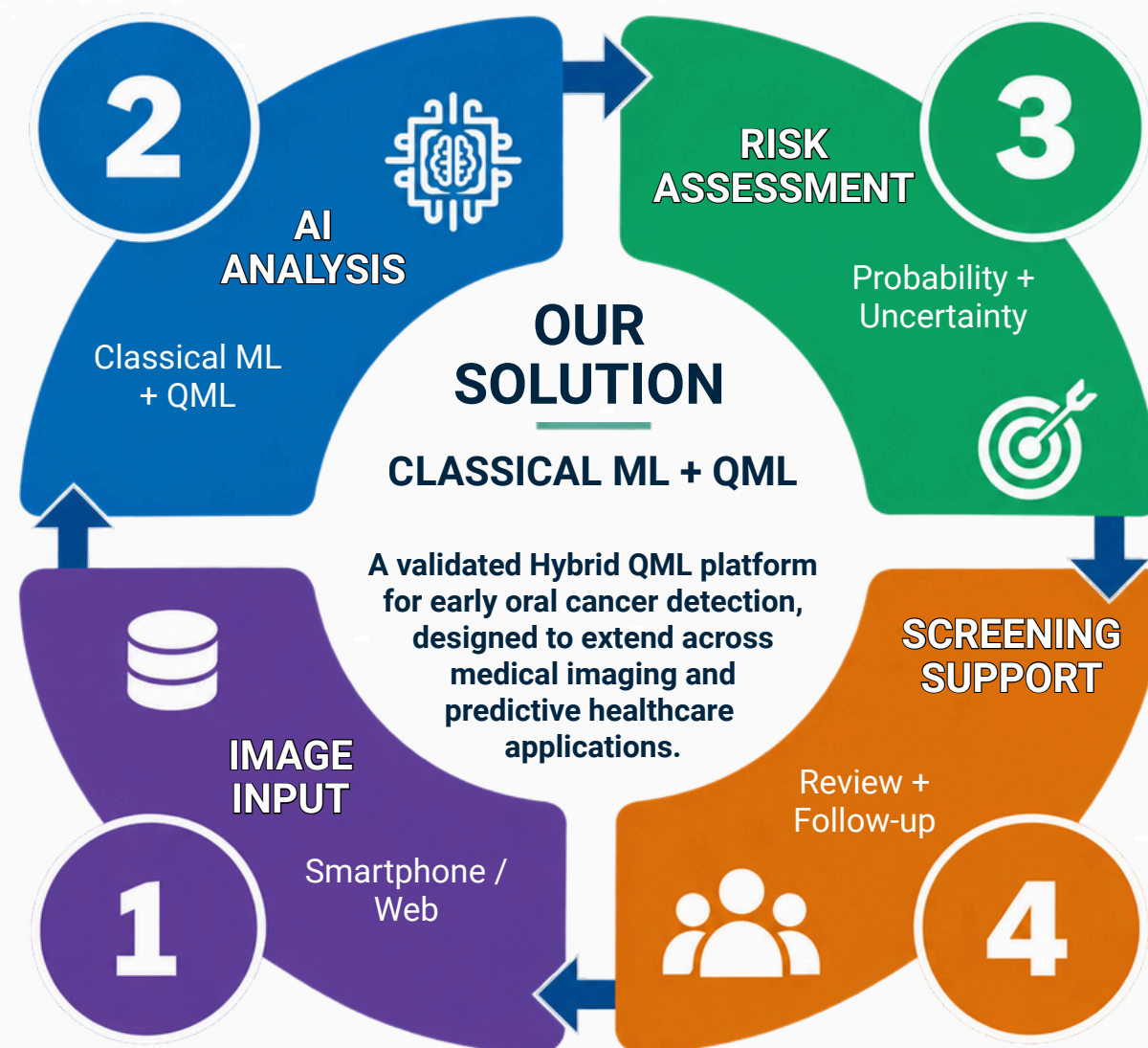


Limited screening
access



High Cost of
treatment

The bottleneck: accessible, consistent, image-based screening.



BEFORE VS AFTER

Visual assessment → AI-assisted pre-screening

Unstructured capture → Guided image acquisition

Manual localization → Automated localization

Specialist-first workflow → Screening + human review

IMPACTS

Earlier Attention | Accessible Screening | 91.3% PR-AUC



HOW IT WORKS



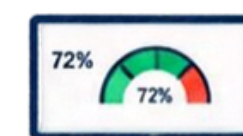
1. Capture (Mobile / Web)



2. Preprocess (Crop & Enhance)



3. Hybrid CNN + QML



4. Risk Probability (+ Uncertainty)



5. Screening & Follow-up

[VIDEO LINK](#)

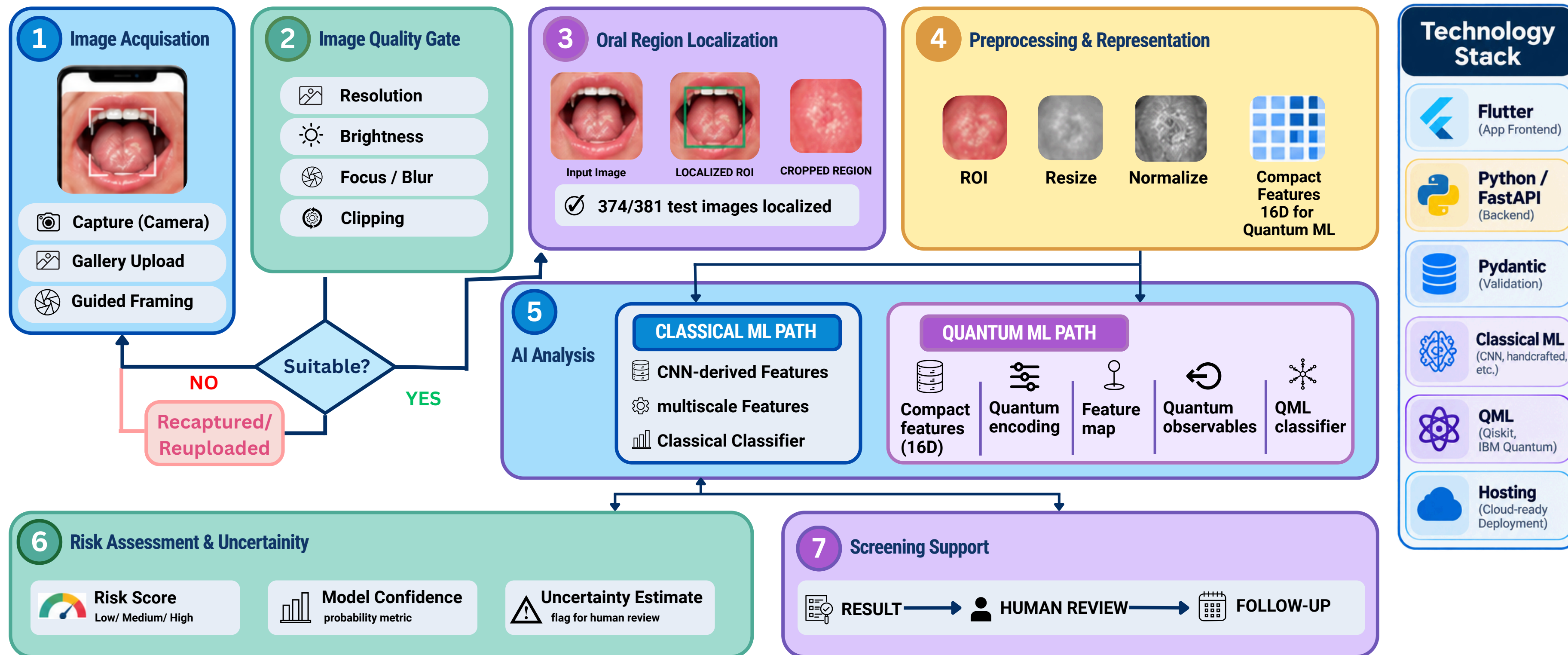


[PROTOTYPE LINK](#)



[REPORT LINK](#)





WHY IT CAN SCALE

Core feasibility pillars that make our solution possible

Technical Feasibility

- Compact feature representation
- Classical + QML pipeline implemented
- Quantum circuits independently verified

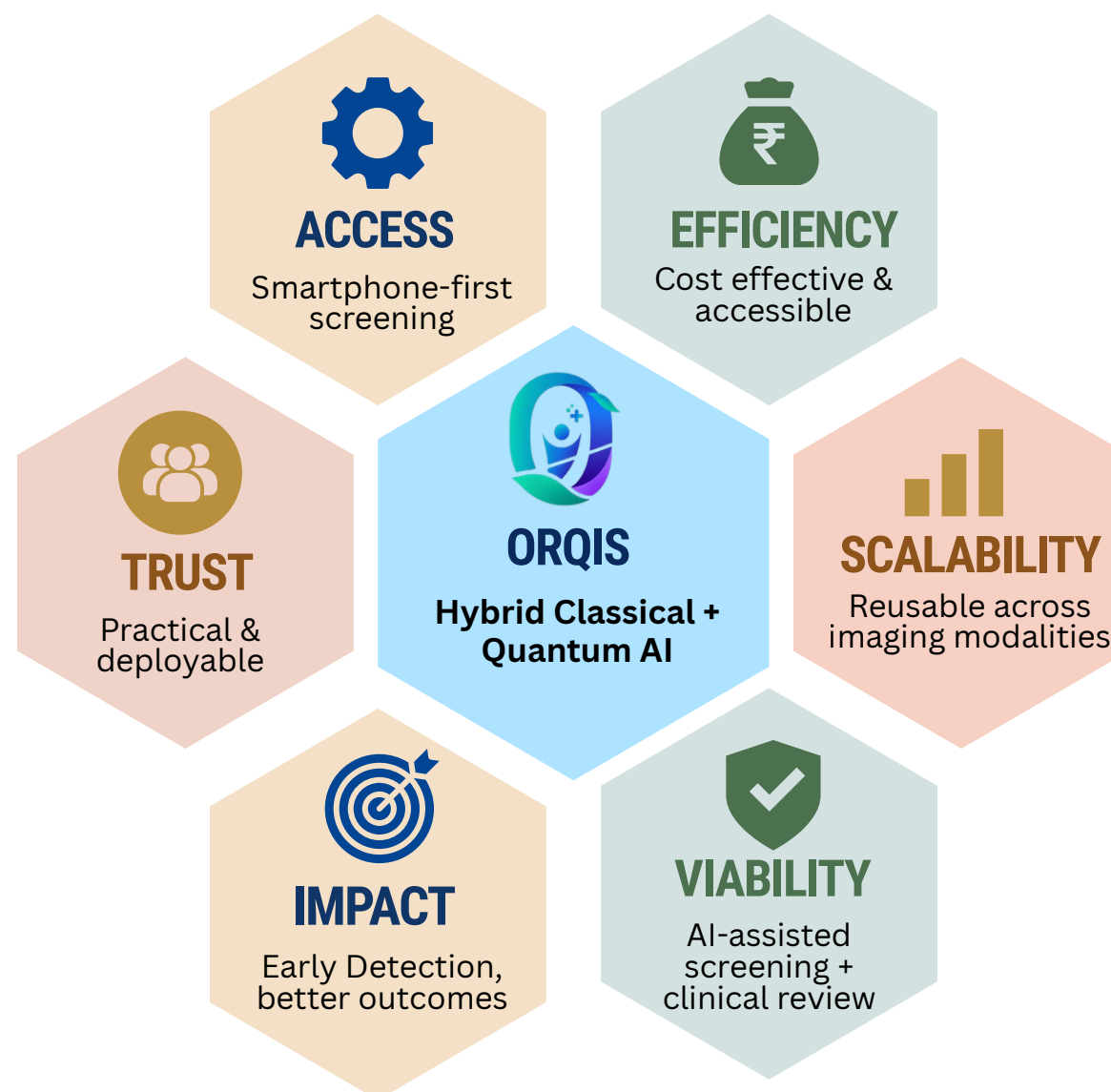
Economic Feasibility

- Commodity smartphone capture
- Open-source ML/QML stack
- No dedicated quantum hardware required

Operational Feasibility

- Automated quality + localization
- Guided acquisition
- Cloud / edge / local deployment paths

BUILDS
THE
FOUNDATION



ENABLES
REAL-WORLD
ADOPTION

IMPACT

From feasibility to scalable & sustainable impact

SCALABILITY

- Modular architecture
- Applicable to new diseases & modalities
- Scales from pilot to national networks

VIABILITY

- Accessible screening (wider reach)
- Uncertainty-aware (safer decisions)
- Sustainable and future-ready

ONE PIPELINE • MULTIPLE MEDICAL MODALITIES

KEY CHALLENGES & OUR SOLUTIONS



Limited labeled data



Patient-level split & augmentation



High-dimensional images



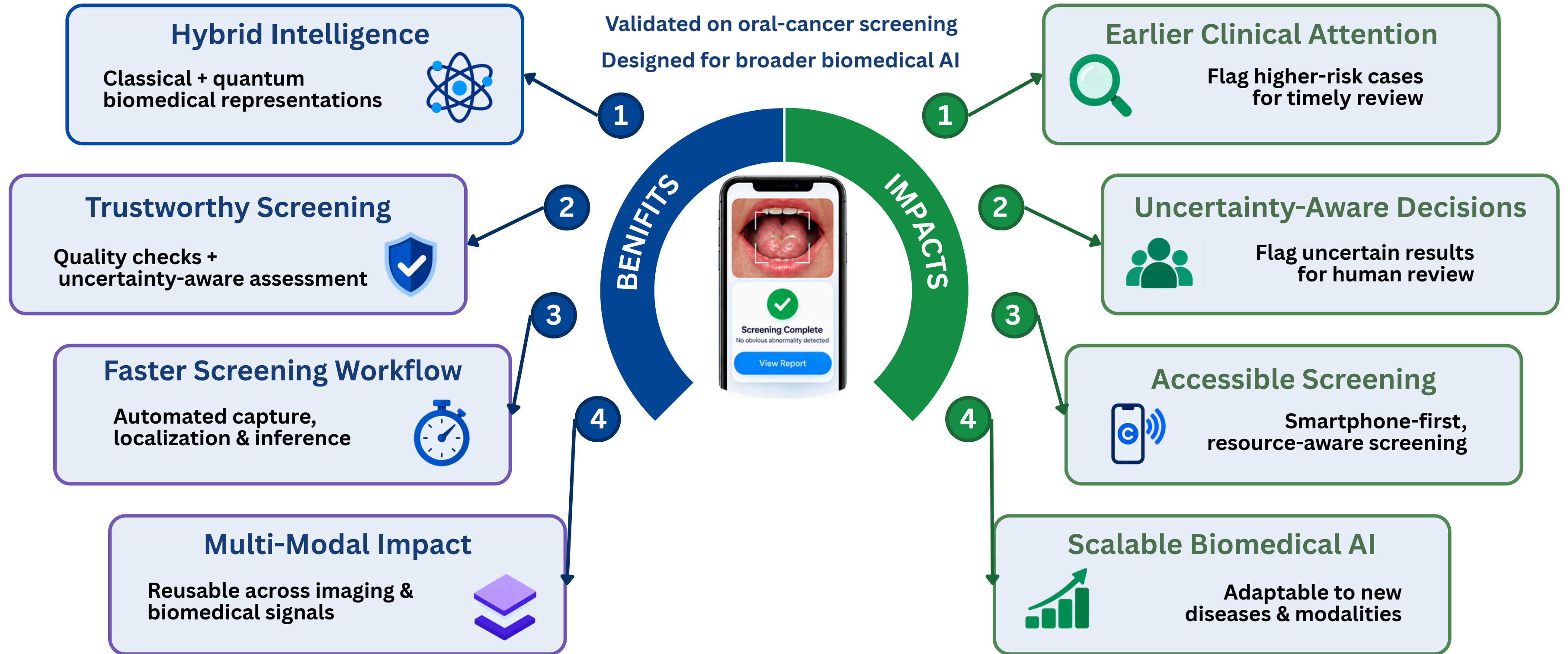
Compact feature representation (16D)



Quantum Resource Cost



Dimensionality reduction before quantum encoding



01

DATASET & EVALUATION

- Peradeniya Oral Cancer Dataset
- ~3,000 images | 714 patients
- 2,436 usable after QC
- Patient-disjoint train / validation / test

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RESEARCH CHALLENGES

- Small, imbalanced medical datasets
- High-dimensional image representations
- Patient-level information leakage risk
- Quantum encoding / resource overhead
- Generalization beyond training patients

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EXPERIMENTAL RESULTS

- 91.3% PR-AUC Best observed result
- More robust on unseen data 89.8% AUROC
- Best in Data Privacy
- Benchmark against existing CNN methods

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BASELINE

- CNN-derived image features
- Handcrafted + multiscale features
- Classical ML controls
- Matched against QML under identical splits

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OUR METHOD

- Quality-controlled image pipeline
- Automated oral-region localization
- Classical ML + quantum representations
- Patient-disjoint evaluation + null testing

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FUTURE SCOPE

- Reusable biomedical pipeline
- Adaptable to MRI, X-ray & other image tasks
- Current validation: oral-cancer imaging
- Wider healthcare applications

KEY REFERENCES

● Datasets
[Link](#)

● Existing Methods
[Link](#)

● Research Gaps
[Link](#)

● Future Scope
[Link](#)

● Experimental Results
[Link](#)